An Introduction to Continuous Random Variables

10 November 2025

Study Guide

This week, we will be learning about continuous random variables. You may find it helpful to list down many examples *and* counterexamples of the various objects we encounter in this week's class.

Definitions

Definition 1. An **continuous random variable** is a random variable that can take on any numeric value within a range of values. In particular, a random variable X is a continuous random variable if there exists a nonnegative function $p(x)$, called a **probability density function**, such that, for any interval of real numbers $[a, b] \subseteq \mathbb{R}$,

$$P(a \leq X \leq b) = \int_a^b p(x)\ dx.$$

Note that the probability density function must satisfy

$$\int_{\mathbb{R}} p(x)\ dx = 1.$$

The **cumulative distribution function** of X is thus given by

$$F(t) = P(X \leq t) = \int_{-\infty}^t p(x)\ dx.$$

Definition 2. Let X be a continuous random variable with probability density function $p(x)$. The **expected value** of X, denoted by $\mathbb{E}[X]$, is given by

$$\mathbb{E}[X] = \int_{\mathbb{R}} x \cdot p(x)\ dx.$$

As with discrete random variables, we can define **variance** and **standard deviation** of X,

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right], \quad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Definition 3. A continuous random variable X is **uniformly distributed** on the interval (a, b) if the probability density function of X is given by

$$p(x) = \begin{cases} \frac{1}{b-a} & \text{when } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 4. A continuous random variable X is **exponentially distributed** with parameter λ if the probability density function of X is given by

$$p(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$